

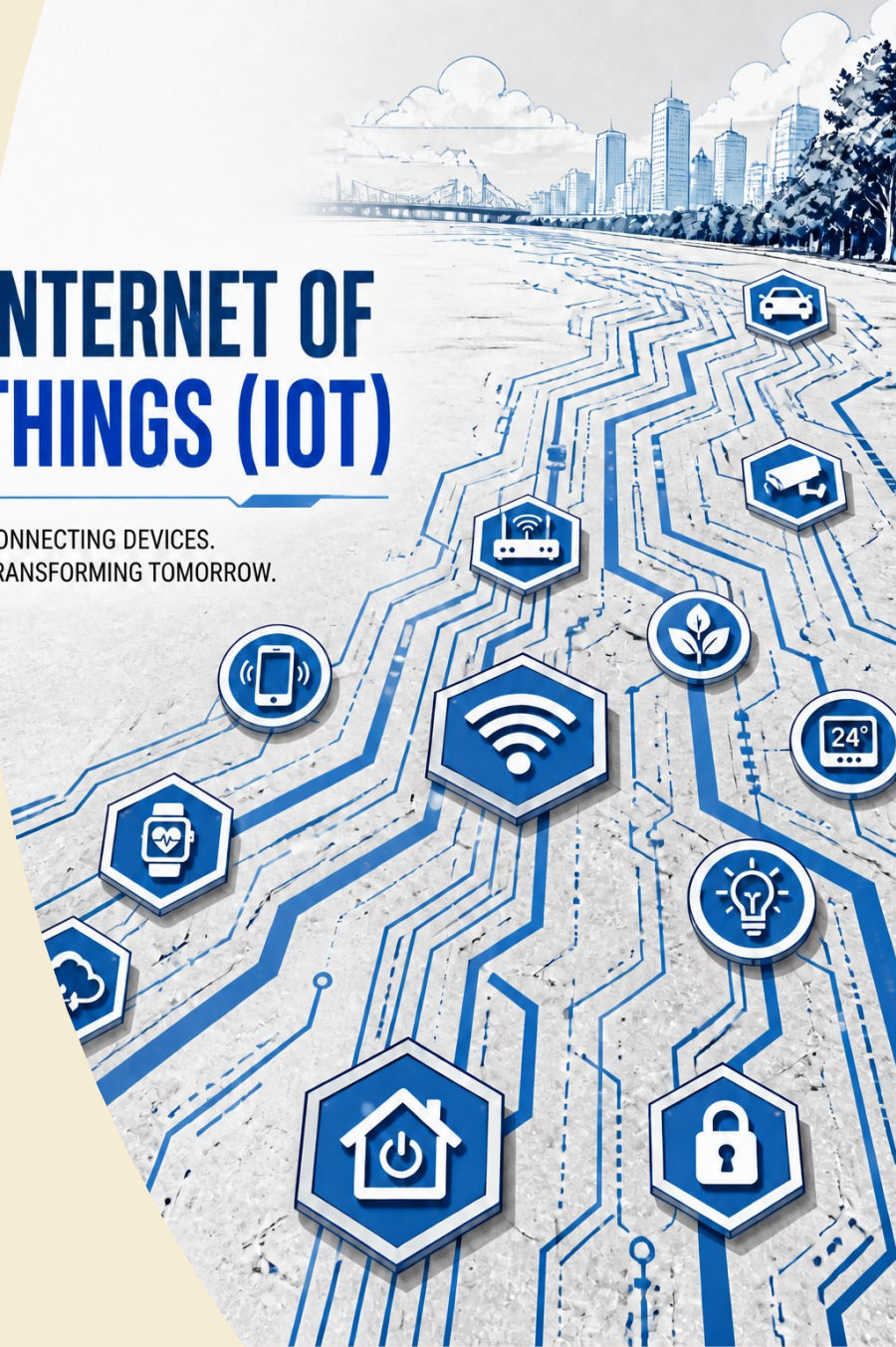
IoT-Based Smart Pothole Detection System

Smart Road Monitoring Using IoT

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INTERNET OF THINGS (IOT)

CONNECTING DEVICES.
TRANSFORMING TOMORROW.



Introduction



What is a pothole?

A surface depression formed by pavement failure — risk to vehicles and people.



Why detect potholes?

Prevents accidents, reduces vehicle damage and lowers long-term repair costs.



IoT in transport

Distributed sensors + connectivity enable continuous, automated road condition monitoring.

Problem Statement

Safety Hazard

Potholes cause accidents and loss of control at speed.

Vehicle Damage

Suspension, tyres and alignment suffer — frequent costly repairs.

Traffic Impact

Unexpected manoeuvres create congestion and delays.

Maintenance Burden

Reactive repairs increase municipal expenditure and slow response times.



Project Objectives



Automatic Detection

Use onboard sensors to detect pothole events in real time.



Location Recording

Capture accurate GPS coordinates for each event.



Cloud Upload

Transmit events to cloud via Wi-Fi (ESP32) for centralised storage.



Authority Dashboard

Visualise incidents, severity and timestamps for maintenance planning.





Hardware Components

- ESP32 microcontroller (sensor aggregation & Wi-Fi)
- HC-SR04 ultrasonic sensor (depth measurement)
- Vibration sensor (impact detection)
- GPS module (location: NMEA data)
- Breadboard, jumper wires, power supply
- LED / Buzzer for local alerts

Software Requirements

- Development

Arduino IDE, Embedded C/C++ for ESP32 firmware

- Connectivity

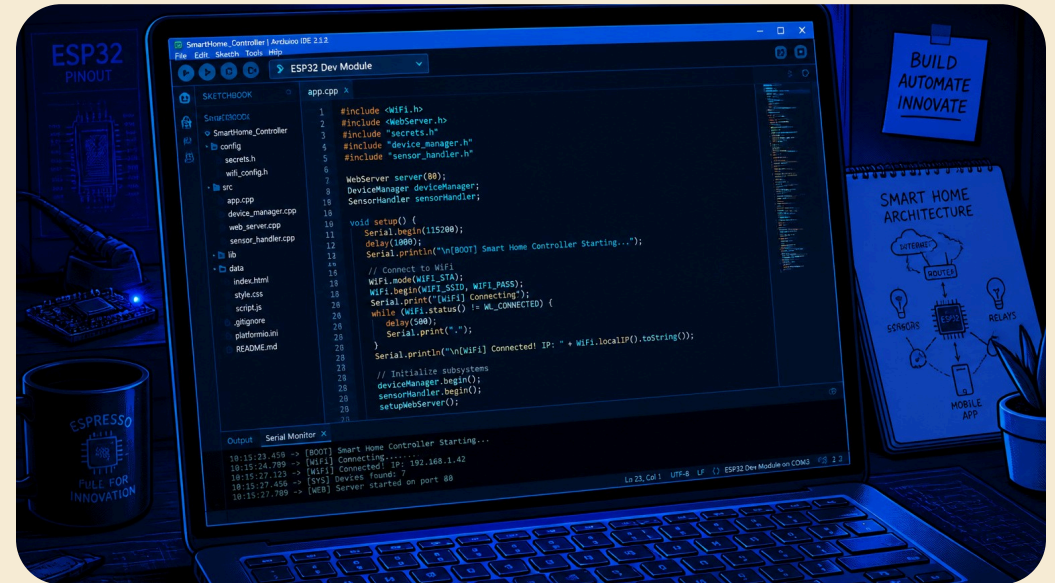
Wi-Fi stack, MQTT/HTTP client for uploads

- Cloud

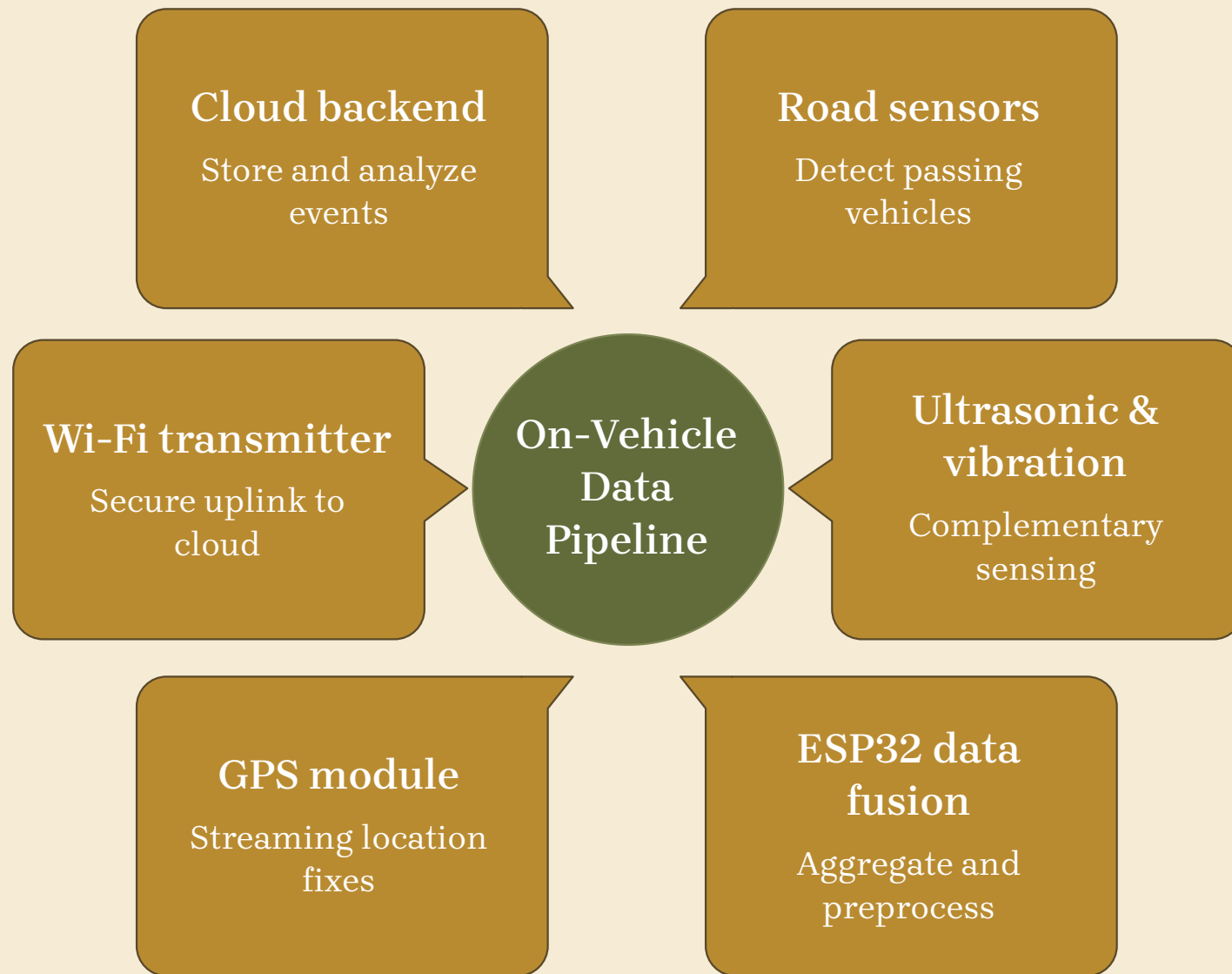
Firebase / ThingSpeak for data ingestion and storage

- Dashboard

Web dashboard (map, markers, severity filters)



System Architecture



The architecture shows on-vehicle sensing and processing, secure upload to cloud storage, and a web dashboard for authorities to review incidents.

Working Principle



1. Detection

Sensors detect depth change and vibration signature consistent with a pothole.



2. Local Processing

ESP32 fuses sensor inputs, filters noise, and validates event thresholds.



3. Geotagging

GPS module provides latitude, longitude and timestamp for the event.



4. Upload

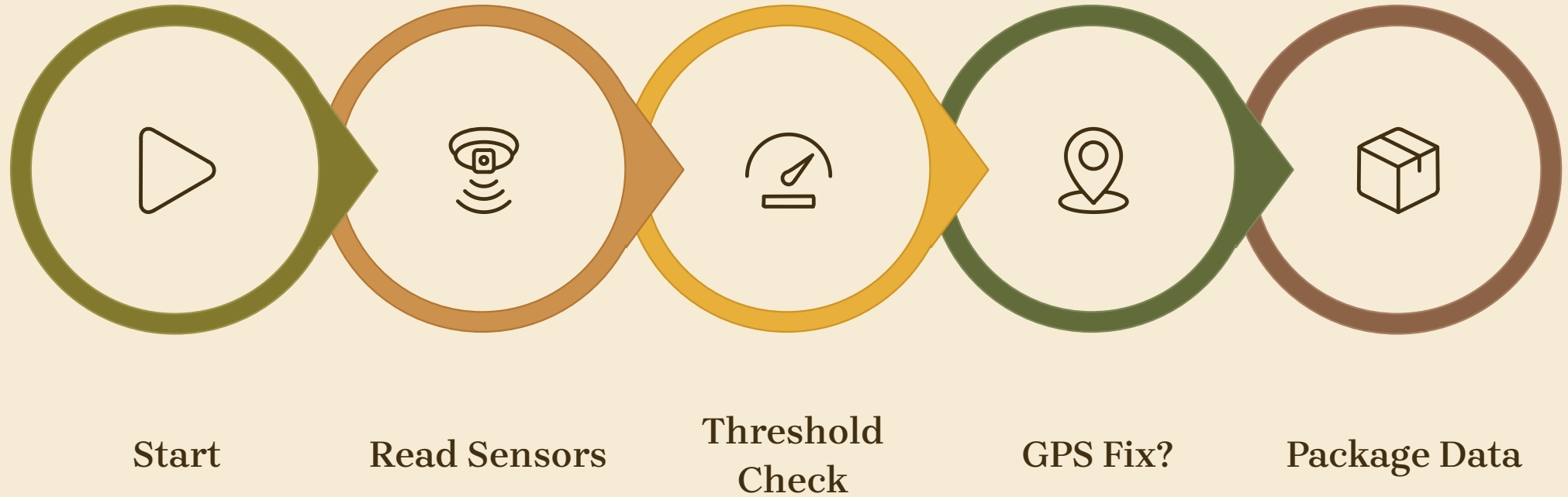
ESP32 sends structured JSON (location, severity, sensor traces) to the cloud.



5. Visualisation

Dashboard displays markers, severity indicators, and allows status updates.

Workflow Flowchart



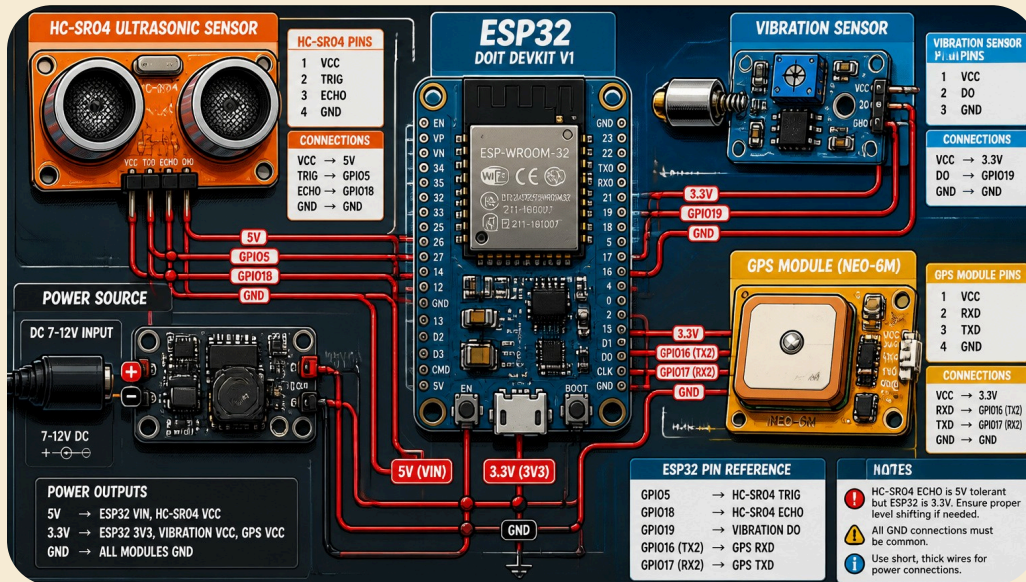
Flowchart includes validation, retry logic and failure handling to ensure reliable event delivery under intermittent connectivity.

Circuit Diagram

Wiring overview:

- HC-SR04: TRIG/TRIG → ESP32 GPIO, ECHO → GPIO (with level shift)
- Vibration sensor → ADC pin of ESP32
- GPS TX/RX → ESP32 RX/TX serial interface
- Power: 5V supply to sensors, 3.3V regulator for ESP32
- LED/Buzzer → GPIO with current-limiting resistor

Follow proper grounding and level-shifting for reliable signal capture.



Thank You

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